

TGTSE: Token-Guided Target Speaker Extraction with Visual Cue

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Abstract

Utilizing visual cues for speaker extraction has demonstrated strong performance in noisy-reverberant environments. This paper proposes a token-guided paradigm for audio-visual target speaker extraction (AVTSE). Given a target speaker's lip movements in a facial video, the first goal is to extract coarse-grained semantic tokens associated with the lip movements. Second, we treat the predicted tokens as temporally aligned semantic cues to guide the speaker extraction system. We evaluate our approach on two benchmark datasets: VoxCeleb2-2Mix and LRS2-2Mix. Evaluation results demonstrate that our token-guided approach consistently outperforms conventional AVTSE systems.

Index Terms: audio-visual target speaker extraction, semantic tokens

1. Introduction

Target speaker extraction (TSE) plays a crucial role in various real-world applications, including online meetings, hearing aid design, and human-robot interaction [1, 2]. Conventional TSE frameworks typically leverage deep neural networks (DNNs) to extract each speaker's voice from an audio mixture by conditioning on an enrollment utterance [3, 4, 5, 6, 7]. While this approach has proven effective in audio-only scenarios, it encounters significant challenges when extended to complex acoustic environments [8]. To name a few, the enrollment utterance is usually not temporally aligned with the mixture, providing only weak global conditioning without fine-grained temporal guidance [5]. A short enrollment segment may lack sufficient contextual information to capture dynamic speaking characteristics such as prosody, articulation timing, and emotional cues [9]. Differently, recent studies have explored leveraging synchronous visual cues (such as lip movements) [10], which are robust against the acoustic environment. Visual signals provide continuous and temporally synchronized contextual information related to the target speaker's speech, which is strongly correlated with the phonetic content and temporal boundaries of speech production [11, 12].

However, directly incorporating raw visual signals into extraction systems is challenging due to modality heterogeneity [13, 14]. Notably, visual signals vary significantly in temporal resolution and frame rate, making direct feature-level fusion or alignment difficult. Therefore, directly conditioning extraction networks on raw visual signals often leads to weak or unstable guidance. To address this, previous studies [15, 11, 16] typically employ a pre-trained face encoder

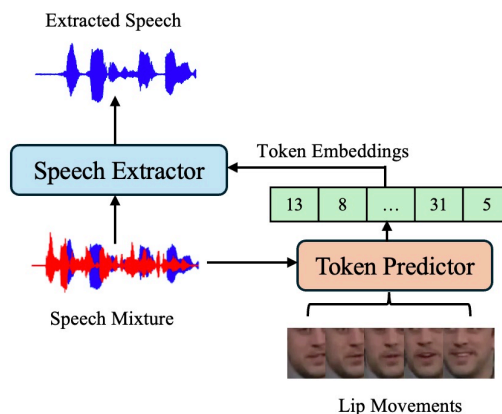


Figure 1: Illustration of token-guided target speaker extraction. Token predictor outputs target speaker's semantic tokens and speech extractor extracts target clean speech.

(e.g., 3D-ResNet18 [13]) to transform each visual frame into a compact lip embedding before conditioning the extraction network. These lip embeddings serve as aligned features that can be more easily fused with audio features, providing additional guidance for TSE [17]. In addition, lip embeddings are related to speaker identity, speech activity, articulation style, and content information. They represent only correlated features, rather than necessary variables for generating the target speech. Moreover, there exists a significant cross-modal gap: the audio mixture resides in the acoustic space, while the lip embeddings reside in the visual space, forcing the network to learn a complex cross-modal alignment mapping.

Compared to continuous lip embeddings, speech tokens derived from self-supervised speech models [18, 19] or neural codecs [20] provide a discrete, causally grounded representation that is more tightly coupled with the speech generation process. Conditioning AVTSE systems on such tokens rather than lip embeddings reduce audio-visual ambiguity and enables the model to focus on speech-content-aware target speech reconstruction rather than relying on audio-visual correlations. To this end, as shown in Fig. 1, we aim to map audio-visual streams into a discrete semantic space, where semantic tokens serve as temporally aligned cues to guide the TSE process.

In this paper, we propose a novel token-guided target speaker extraction (TGTSE) framework that introduces discrete tokens as semantic clues for audio-visual TSE (AVTSE). The key idea is that discrete semantic tokens provide a modality-agnostic, semantically-meaningful, and temporally-consistent representation that bridges the visual modality with the acoustic domain. This design is particularly advantageous for the following reasons: ① discrete tokens exhibit strong temporal

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synchrony with speech, greatly simplifying alignment; ② tokens provide the explicit semantic structure that raw visual embeddings lack; ③ token prediction converts complex visual-to-audio mappings into a sequence modeling problem that can be robustly trained using techniques from speech recognition and audio generation.

2. Token-Guided Target Speaker Extraction

2.1. Task Formulation

Let y denote a monaural audio mixture consisting of the target speaker’s clean speech s and non-target interfering signals n :

$$y = s + n \in \mathbb{R}^N, \quad (1)$$

with N denoting the signal length. In audio-visual scenes, given a facial video stream with synchronized¹ lip movements $v \in \mathbb{R}^{N_v \times 3 \times h \times w}$, we aim to predict discrete semantic tokens that strongly related to the video stream. This is accomplished using a token predictor (denoted as DNN_1):

$$\ell = \text{DNN}_1(y, v) \in \mathbb{R}^{M \times K}, \quad (2)$$

where M is the number of tokens and K is token categories. $\ell \in \mathbb{R}^{M \times K}$ denotes logits (i.e., unnormalized scores) for each token at every time frame. The predicted semantic tokens $\hat{z}$ is then obtained via:

$$\hat{z} = \arg \max_{k \in \{1, \dots, K\}} \text{softmax}(\ell_k) \quad (3)$$

We train DNN_1 via supervised learning based on pseudo token labels $z \in \{1, \dots, K\}^M$ produced by WavLM-KM [21]. In detail, frame-level representations² are extracted using a frozen self-supervised learning model (i.e., WavLM [18]). K-means clustering is applied to all representations to obtain K cluster centroids, and each frame in s is assigned to its nearest centroid, producing pseudo token labels z . Following the previous study [20], we use a subset of LibriSpeech³ to reimplement WavLM-KM. In this paper, we use WavLM-Large⁴, yielding about 50 frames per second at 16 kHz sample rate (approximately 20 ms per frame). Token prediction is formulated as a frame-level classification problem, with the cross-entropy loss defined as:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{M} \sum_{m=1}^M \log \frac{\exp(\ell_{m,z_m})}{\sum_{k=1}^K \exp(\ell_{m,k})}, \quad (4)$$

where $\ell_{m,k}$ is the logit for the m -th time frame at the k -th token category, and z_m is the m -th token in z . Subsequently, the predicted tokens $\hat{z}$ serve as temporally-aligned semantic cues that condition a token-guided target speaker extractor (denoted as DNN_2). Each token is mapped to a continuous embedding vector using a trainable codebook:

$$e_m = E_{\hat{z}_m}, \text{ where } E \in \mathbb{R}^{K \times D}, \quad (5)$$

¹Although v and s may have different length ($N \neq N_v$), they are assumed to have the same overall time duration.

²Note: we collect all layer-wise hidden states and compute their average to obtain the final feature embeddings.

³<https://openslr.ttml.net/resources/12/train-clean-100.tar.gz>.

⁴<https://huggingface.co/microsoft/wavlm-large>.

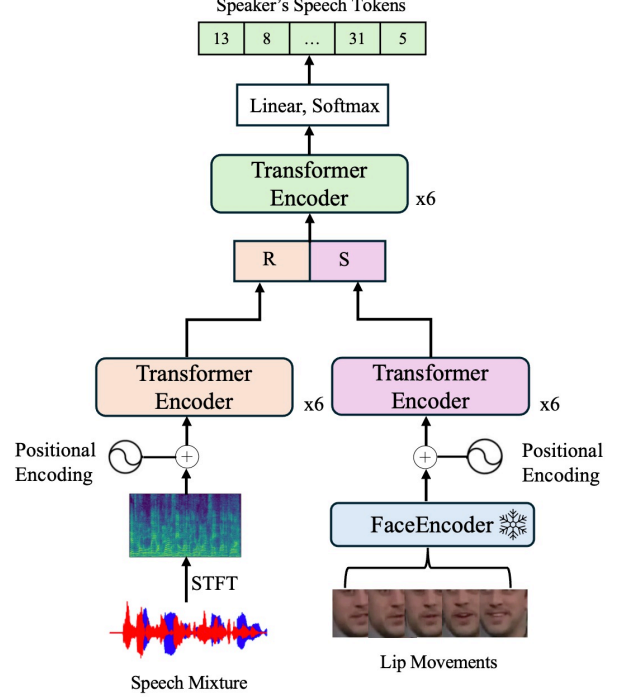


Figure 2: Illustration of token predictor. Best viewed in color.

where e_m is the m -th token embedding, E the trainable token embedding matrix, and D the embedding dimension. The resulting token embeddings $e \in \mathbb{R}^{M \times D}$ are then used to condition the speaker extractor:

$$\hat{s} = \text{DNN}_2(y, e), \quad (6)$$

DNN_2 and the token embeddings are trained end-to-end to minimize the reconstruction error between the DNN estimate $\hat{s}$ and clean target s , using the negative scale-invariant signal-to-distortion ratio (SI-SDR) loss [22].

In the following subsections, we detail the architectures of the token predictor and token-guided target speaker extractor.

2.2. Token Predictor

Figure 2 illustrates the token predictor. A frozen face encoder pre-trained on lip-reading task (e.g., 3D-ResNet18 [13]) is employed to extract frame-level visual features. To match the temporal resolution of tokens, an upsampling layer interpolates the features along the time axis, i.e.,

$$V = \text{FaceEncoder}(v) \in \mathbb{R}^{N_v \times H}, \quad (7)$$

$$S = \text{Upsample}(V) \in \mathbb{R}^{M \times H}, \quad (8)$$

where H denotes the feature dimension of each visual frame.

To enhance the accuracy of token prediction, we also use mixture as the input to the token predictor. Following previous work [13], we transform the mixture y into 321-dimensional spectral magnitudes $|\text{STFT}(y)| \in \mathbb{R}^{T_{\text{mix}} \times 321}$ via short-time Fourier transform (STFT) with a 40 ms window and 10 ms hop-length at 16 kHz sample rate. Since each token corresponds to approximately 20 ms of speech (as extracted by WavLM-Large), each token spans two STFT frames. As a result, the total number of STFT frames satisfies $T_{\text{mix}} = 2M$. To align

the STFT frames with the token sequence, we apply a Conv1D layer with kernel size 2, stride 2, and H output channels:

$$R = \text{Conv1D}(|\text{STFT}(y)|) \in \mathbb{R}^{M \times H}, \quad (9)$$

After that, we use a stack of transformer encoder [23] layers with sinusoidal positional encodings [23] to model S and R , i.e.,

$$\begin{aligned} \tilde{S} &= \text{TransformerEncoder}(\text{PE}(S)) \in \mathbb{R}^{M \times H}, \\ \tilde{R} &= \text{TransformerEncoder}(\text{PE}(R)) \in \mathbb{R}^{M \times H}, \\ X &= \text{TransformerEncoder}(\text{Conv1D}([\tilde{S}, \tilde{R}])) \in \mathbb{R}^{M \times H}, \end{aligned} \quad (10)$$

Finally, a linear classification head produces the logits:

$$\ell = WX + b, \quad (11)$$

where $W \in \mathbb{R}^{H \times K}$ and $b \in \mathbb{R}^K$ are the learnable weights and bias of the linear classification layer that maps each frame-level feature to probability.

2.3. Token-Guided Speaker Extractor

To investigate the effectiveness of the token-guided approach, we use time-frequency (T-F) and time-domain speech extraction networks.

2.3.1. Time-Frequency Domain Speaker Extractor

We first transform the mixture waveform y into its complex spectrogram representation $Y = \text{STFT}(y) \in \mathbb{R}^{T \times F}$, where F denotes the number of frequency bins. A convolutional encoder is then applied to map the real and imaginary (RI) components into a latent representation:

$$g = \text{Conv2D}([\mathcal{R}(Y), \mathcal{I}(Y)]) \in \mathbb{R}^{Q \times T \times F}. \quad (12)$$

where $\mathcal{R}(\cdot)$ and $\mathcal{I}(\cdot)$ respectively extract RI components. Q denotes the embedding dimension of each T-F unit. To incorporate token embeddings $e \in \mathbb{R}^{1 \times M \times D}$, two linear layers respectively transform the feature dimension D to F and 1 to Q . Then, we upsample the token sequence from M frames to T frames. These steps result in $\tilde{e} \in \mathbb{R}^{Q \times T \times F}$. We fuse the token embeddings with the mixture representation by concatenation and a linear projection:

$$\tilde{g} = \text{Linear}([\tilde{e}, g]) \in \mathbb{R}^{Q \times T \times F}, \quad (13)$$

which is processed by a stack of T-F modeling blocks (e.g., TF-GridNet [24]) to refine the mixture representation:

$$G = \text{TF-Blocks}(\tilde{g}) \in \mathbb{R}^{Q \times T \times F}. \quad (14)$$

Finally, a deconvolutional decoder maps G back to complex spectrogram and iSTFT reconstructs target speaker's speech:

$$\hat{s} = \text{iSTFT}(\text{DeConv2D}(G)) \in \mathbb{R}^N. \quad (15)$$

2.3.2. Time-Domain Speaker Extractor

Time-domain network consists of three main components: a convolutional encoder, a masking-based estimator, and a linear decoder with overlap-and-add reconstruction. Formally, the processing pipeline is:

$$Z = \text{ReLU}(\text{Conv1D}(y)) \in \mathbb{R}^{M \times L}, \quad (16)$$

$$\mathcal{M} = \mathcal{E}(Z, e) \in \mathbb{R}^{M \times L}, \quad (17)$$

$$\hat{Z} = Z \otimes \mathcal{M} \in \mathbb{R}^{M \times L}, \quad (18)$$

$$\hat{s} = \text{Overlap-Add}(\text{Linear}(\hat{Z})) \in \mathbb{R}^N, \quad (19)$$

where Z is an M -frame embedding sequence of dimension L , and $\otimes$ is element-wise multiplication. The mask-based estimator $\mathcal{E}(\cdot, \cdot)$ incorporates the token embeddings into the mixture embeddings and produces a multiplicative mask $\mathcal{M}$ that selectively enhances the components of target speaker while attenuating interfering sources. Finally, the decoder reconstructs the time-domain waveform from the masked features $\hat{Z}$ via a linear layer followed by the standard overlap-add synthesis [25], producing the estimate $\hat{s}$.

3. Experimental Setup

To validate TGTSE, we use two benchmarks, VoxCeleb2-2Mix and LRS2-2Mix, which are respectively derived from VoxCeleb2 [32] and LRS2 [13]. Both datasets are constructed using the same configuration following the pre-processing pipeline in ClearerVoice [31], in which 20,000, 5,000, and 3,000 two-speaker mixtures are respectively simulated for training, validation and testing, and there is no speaker overlap across all partitions. The target speech is mixed with interfering speech at a signal-to-noise ratio (SNR) randomly sampled from -10 to $+10$ dB. All audio signals are sampled at 16 kHz, while video signals are processed at 25 frames per second, with each frame converted to a $1 \times 112 \times 112$ grayscale image. Each of the training and validation samples consists of 3-second speech segments, whereas test samples use full-length utterances. Furthermore, we evaluate the generalization capability on three cross-domain datasets: LRS3 [33], Grid [34] and TCD-TIMIT [35]. For the token predictor, we report its accuracy at token prediction. In addition, we evaluate the AVTSE performance using four commonly adopted metrics: signal-to-distortion ratio (SDR) [36], scale-invariant SDR (SI-SDR) [22], perceptual evaluation of speech quality (PESQ) [37], and short-time objective intelligibility (STOI) [38].

For the token predictor, each transformer encoder consists of 6 layers with a hidden size of 256, a feed-forward dimension of 1024, and 8 self-attention heads (totaling 25.4M parameters). We optimize the network using AdamW [39] starting with a learning rate of 1×10^{-4} , and employ a cosine learning rate schedule with linear warmup during the first 10% training epochs, and then decays following a cosine function for the remaining epochs. For speaker extraction, we employ two extraction networks: TFGridNet [30] and MossFormer2 [31], which are state-of-art networks respectively in the T-F and time domain. Following the symbols defined in their original papers, we set the hyper-parameters of TFGridNet to $D = 128, B = 6, I = 1, J = 1, H = 256, L = 4, E = 8$ (resulting in 11.3M parameters), and those of MossFormer2 to $R = 24, N = 512, K = 16, N' = 256, L = 2$ (leading to 55.7M parameters). Please do not confuse these symbols with the ones defined in this paper. The models are trained for up to 50 epochs. For TFGridNet, the learning rate is set to 5×10^{-4} with Adam [40], which is decayed by a factor of 0.5 if the validation loss does not improve for two consecutive epochs. For MossFormer2, we use the same training strategy as the token predictor.

4. Evaluation Results

To investigate the impact of different token categories, we condition TFGridNet on pseudo token labels from WavLM-KM and report the performance of the codebook size K ranging from 32 to 1024 in Figure 3a, we observe that the best performance is $K = 128$, and increasing K does not lead to im-

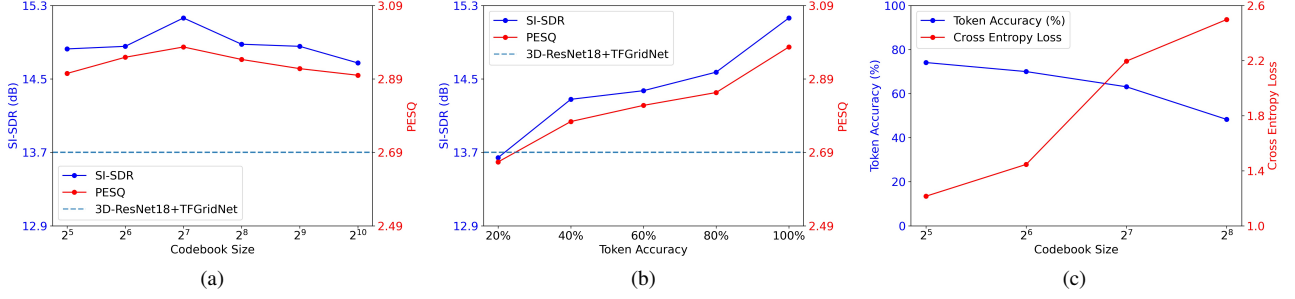


Figure 3: Ablation study on VoxCeleb2-2Mix when using TFGridNet for TGTSE: (a) oracle performance of speaker extractor using codebook size from 32 to 1024; (b) performance of speaker extractor with different token accuracy from 20% to 100%; and (c) accuracy of token predictor using codebook size from 32 to 256.

Table 1: Comparison with various AVTSE networks on VoxCeleb2-2Mix and LRS2-2Mix. “–” means that number is not reported in the original paper.

System	Year	#Params (M)	Domain	VoxCeleb2-2Mix				LRS2-2Mix			
				SDR↑	SI-SDR↑	PESQ↑	STOI↑	SDR↑	SI-SDR↑	PESQ↑	STOI↑
Unprocessed	–	–	–	0.0	–0.1	1.24	0.63	0.2	0.0	1.18	0.66
VisualVoice [10]	2021	77.8	T-F	10.2	9.2	1.97	–	12.0	11.5	–	–
AV-ConvTasNet [15]	2019	20.2	T	10.9	10.5	2.07	0.85	12.1	11.6	2.11	0.90
AV-DPRNN [26]	2022	15.3	T	11.8	11.4	2.07	0.87	12.6	12.0	1.99	0.90
MuSE [11]	2021	25.0	T	12.0	11.6	2.21	0.86	11.6	11.0	1.98	0.90
AV-SepFormer [16]	2023	21.7	T	12.5	12.1	2.25	0.88	13.2	12.6	2.20	0.91
reentry [27]	2022	18.8	T	12.8	12.5	2.34	0.88	–	–	–	–
SEANet [28]	2025	8.7	T	13.3	12.9	2.34	0.89	13.7	13.0	2.25	0.92
Uni-Net [8]	2024	5.0	T-F	13.8	13.5	2.66	0.91	–	–	–	–
AV-CrossNet [29]	2025	11.1	T-F	–	14.7	2.93	–	–	–	–	–
AV-TFGridNet [30]	2023	20.8	T-F	14.1	13.6	2.69	0.91	15.7	15.1	2.62	0.94
AV-MossFormer2 [31]	2025	68.5	T	14.9	14.5	2.66	0.90	16.0	15.5	2.61	0.94
TGTSE (TFGridNet)	2026	11.3 + 25.4	T-F	14.8	14.5	2.83	0.92	15.9	15.3	2.77	0.95
TGTSE (MossFormer2)	2026	55.7 + 25.4	T	15.2	14.8	2.74	0.92	16.2	15.7	2.68	0.95

Table 2: Cross-domain evaluation across various AVTSE networks on the LRS3, TCD-TIMIT and Grid datasets. All systems are trained on VoxCeleb2-2Mix.

Dataset	System	SDR (dB)↑	SI-SDR (dB)↑	PESQ↑	STOI↑
LRS3	Unprocessed	0.1	0.0	1.28	0.69
	AV-TFGridNet [30]	17.0	16.8	2.94	0.94
	AV-MossFormer2 [31]	17.5	17.2	2.82	0.94
	TGTSE (TFGridNet)	17.9	17.7	3.14	0.95
	TGTSE (MossFormer2)	18.1	17.8	3.07	0.95
TCD-TIMIT	Unprocessed	0.2	0.0	1.45	0.68
	AV-TFGridNet [30]	17.5	17.0	3.05	0.94
	AV-MossFormer2 [31]	18.3	17.9	2.77	0.93
	TGTSE (TFGridNet)	18.3	17.9	3.43	0.96
	TGTSE (MossFormer2)	19.6	19.2	3.06	0.95
Grid	Unprocessed	–0.3	–0.5	1.35	0.67
	AV-TFGridNet [30]	15.1	13.2	2.83	0.90
	AV-MossFormer2 [31]	15.3	13.6	2.68	0.90
	TGTSE (TFGridNet)	17.7	16.2	3.14	0.94
	TGTSE (MossFormer2)	17.4	16.1	2.88	0.94

provement. Regardless of the category of tokens, oracle performance is superior to 3D-ResNet18+TFGridNet, demonstrating the feasibility of using semantic tokens for conditioning. In Figure 3b, we randomly replace part of the pseudo labels to achieve token accuracy ranging from 20% to 100%, and then train TGTSE models. It can be seen that with an accuracy rate of 60%, TGTSE already has reasonable performance. In Figure 3c, we further set K from 32 to 256 for training token pre-

dictor, the accuracy is decreasing, and the cross-entropy loss is difficult to further decrease. This means that the larger the number of token categories, the more difficult it is to make accurate token prediction.

Based on the above observations, we choose to use the token predictor version with K set to 128 for further training the speaker extraction network, and report the extraction performance on two benchmark datasets in Table 1. Evaluation results demonstrate a consistent trend of performance improvement in recent AVTSE networks. TGTSE (MossFormer2) achieves the highest SDR of 15.2/16.2 dB and SI-SDR of 14.8/15.7 dB respectively for the VoxCeleb2-2Mix and LRS2-2Mix datasets. Compared to AV-TFGridNet, our TGTSE (TFGridNet) improves SDR by 0.7 dB and SI-SDR by 0.9 dB.

In Table 2, we randomly generate 3,000 speech mixtures from LRS3, TCD-TIMIT and Grid for cross-domain evaluation with the SNR ranging from -10 to 10 dB. To ensure a fair comparison, we use the checkpoints from the ClearerVoice repository as the baselines, our TGTSE performs the best across the three datasets.

5. Conclusions

We have proposed TGTSE, a token-guided TSE approach leveraging visual cues for speaker conditioning. It converts raw visual signals to semantic tokens to improve target-speaker extraction. Evaluation results on two representative benchmark datasets show that TGTSE consistently outperforms existing AVTSE networks.

6. Generative AI Use Disclosure

The authors used ChatGPT as a language assistance tool, including clarity improvement and grammar correction. All scientific content, analyses, and conclusions were developed and verified by the authors, who take full responsibility for the work.

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